Why a DNA Chain Has Many Possible Shapes

Microstates, multiplicity, entropy, and stretching

Use a simple flexible-chain model to explore how many different molecular configurations can produce the same overall displacement. Begin with the three-segment example from class, then use AI-assisted coding to investigate longer chains and the effect of a pulling force.

Model scope: This is a one-dimensional teaching model inspired by a flexible molecular chain. It does not describe DNA base pairing, its full three-dimensional structure, or sequence-dependent folding. Segments may overlap, and every configuration has the same energy when no force is applied.

1 Count the possible shapes

Represent a chain as three connected segments. Each segment points either right (+1) or left (−1). Let a be the length of one segment. Fix the starting endpoint at zero.

$$X = a(s_1 + s_2 + s_3)$$

Here X is signed endpoint displacement, and each s_i is +1 or −1. For this practice, use a = 1 arbitrary length unit.

- The configuration (+1, +1, +1) has X = 3a.
- The configuration (+1, +1, −1) has X = a.

Answer before coding:

- List all eight configurations and calculate X for each.
- Group the configurations by X = −3a, −a, a, and 3a. How many belong to each group?
- If all configurations are equally likely, which displacements are most likely?
- Repeat the grouping using end-to-end distance |X|. Explain why its counts differ from the counts for signed displacement.

Coding activity: Generate all eight configurations, draw their segment sequences, and display a bar chart of the counts for each X. Label any vertical offsets used in the drawing as visual aids, not an extra physical dimension.

Check: The counts must add to eight, and the probabilities must add to one.

What to keep in mind

A microstate is one complete sequence of segment directions. Multiplicity Ω(X) is the number of microstates that share the same displacement. Several different shapes can therefore have the same measured X.

2 Explore longer chains

Question: Does a longer chain usually become fully stretched?

$$X = a \sum_{i=1} s_i$$

Simulate chains with N = 10, 50, and 100 segments, using a = 1. Independently choose each segment's direction with equal probability. Generate 10,000 chains for each length; use random sampling rather than listing all configurations.

- Plot the distribution of signed endpoint displacements for each N.
- Calculate the fraction that are fully extended, meaning $|X| = Na$.
- Calculate the root-mean-square distance, $\sqrt{\langle X^2 \rangle}$, and compare it with the total chain length Na.
- Explain why having more segments does not mean that a chain is usually straight.

Validation checks for independent, equally likely directions:

$$\langle X \rangle \approx 0 \quad \text{and} \quad \langle X^2 \rangle \approx Na^2$$

A mean displacement near zero does not mean every chain's endpoints meet. Positive and negative displacements cancel in the average.

Rare-event check: The exact probability of being fully extended is $2/2^N$. If your simulation finds no fully extended chains, explain why this does not prove that the probability is zero.

3 Connect the shapes to entropy

Question: Why are some overall displacements more probable than others?

For N = 10, enumerate all $2^{10} = 1{,}024$ configurations. Count Ω(X), the number with each allowed displacement, and calculate:

$$S(X) = k_B \ln \Omega(X)$$

$$P(X) = \Omega(X)/2^N$$

Here k_B is Boltzmann's constant, and ln is the natural logarithm. Plot Ω(X) and $S(X)/k_B$ against X/a. Calculate entropy only for allowed displacements with nonzero counts.

- Which displacement has the greatest multiplicity and entropy?
- How does the entropy of a fully extended chain compare?
- Explain how many possible arrangements can make a displacement more probable, even when all individual microstates have equal energy.

Check: Summing Ω(X) over all allowed displacements must give 1,024. Compare the exact probability distribution with the sampled N = 10 results from Part 2.

4 Optional challenge on pulling the chain

Question: How does an external pulling force change the chain's configurations?

Let a force f pull the endpoint to the right. Assign each configuration an energy E = −fX. A positive displacement then has a lower energy under a positive force.

$$P(X) = \Omega(X) \exp[fX/(k_B T)] / Z$$

$$Z = \Sigma \text{ over allowed } X' \text{ of } \Omega(X') \exp[fX'/(k_B T)]$$

Z is the normalization factor that makes the probabilities sum to one. T is absolute temperature. Multiplicity favors displacements with many arrangements, while the pulling force favors positive extension.

Use the dimensionless pulling strength $q = fa/(k_B T)$. For N = 10 and a = 1, compare q = 0, 0.25, 0.5, 1, and 2. You can use exp(qX/a) directly, without selecting physical force or temperature values.

- Plot the displacement distribution for each q.
- Calculate $\langle X \rangle = \Sigma\, X\, P(X)$, then plot the mean fractional extension $\langle X \rangle/(Na)$ against q.
- Explain why pulling can favor extended configurations even though fewer configurations are fully extended.
- Check that q = 0 recovers the original distribution and that $|\langle X \rangle|$ never exceeds Na.

Additional mathematical check for this particular model:

$$\langle X \rangle/(Na) = \tanh(q)$$

Optional extension: Try negative q. Predict the direction of the mean displacement before running the code.

5 Coding support

You may use this starter prompt in Colab Gemini or another coding assistant:

Build a Python simulation of a simplified one-dimensional flexible chain. Each segment points left or right. Show all eight configurations of a three-segment chain and group them by signed endpoint displacement. Distinguish signed displacement X from distance |X|. Then sample 10,000 chains for N = 10, 50, and 100, plot their distributions, and check the supplied mean and mean-square formulas. Enumerate all configurations only for small N. For N = 10, calculate multiplicity and entropy. Explain the code, label units, and include checks that counts and probabilities are correct.

For the challenge, ask the coding assistant to add the force-dependent Boltzmann probabilities and compare the computed mean extension with tanh(q).

Reflection

Prepare your configuration counts, comparison plots, and a short explanation connecting microstates, multiplicity, entropy, and probability. Identify two ways that this simplified chain differs from real DNA. Parts 1–3 are the main optional practice; Part 4 is an additional challenge.